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SELECTIVE ETCHING OF ALTERNATING LAYERS OF SILICON OXIDE AND SILICON NITRIDE FOR HIGH ASPECT RATIO CONTACTS

Abstract

Exemplary semiconductor processing methods may include flowing a fluorine-containing precursor into a processing region of a semiconductor processing chamber. A substrate may be positioned within the processing region. The substrate may include alternating layers of silicon nitride and silicon oxide. The methods may include forming plasma effluents of the fluorine-containing precursor. The methods may include contacting the substrate with the plasma effluents of the fluorine-containing precursor. The contacting may selectively etch an exposed portion of silicon nitride. The methods may include introducing a phosphorous-and-fluorine-containing precursor into the processing region of the semiconductor processing chamber while maintaining a flow of the fluorine-containing precursor. The methods may include forming plasma effluents of the fluorine-containing precursor and the phosphorous-and-fluorine-containing precursor. The methods may include contacting the substrate with the plasma effluents of the fluorine-containing precursor and the phosphorous-and-fluorine-containing precursor. The contacting may selectively etch an exposed portion of silicon oxide.

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Background/Summary

TECHNICAL FIELD

[0001] The present technology relates to semiconductor processes and equipment. More specifically, the present technology relates to cyclically etching silicon oxide material and silicon nitride material.

BACKGROUND

[0002] Integrated circuits are made possible by processes which produce intricately patterned material layers on substrate surfaces. Producing patterned material on a substrate requires controlled methods for removal of exposed material. Chemical etching is used for a variety of purposes including transferring a pattern in photoresist into underlying layers, thinning layers, or thinning lateral dimensions of features already present on the surface. Often it is desirable to have an etch process that etches one material faster than another facilitating, for example, a pattern transfer process. Such an etch process is said to be selective to the first material. As a result of the diversity of materials, circuits, and processes, etch processes have been developed with a selectivity towards a variety of materials.

[0003] Etch processes may be termed wet or dry based on the materials used in the process. A wet HF etch preferentially removes silicon oxide over other dielectrics and materials. However, wet processes may have difficulty penetrating some constrained trenches and also may sometimes deform the remaining material. Dry etches produced in local plasmas formed within the substrate processing region can penetrate more constrained trenches and exhibit less deformation of delicate remaining structures. However, local plasmas may damage the substrate through the production of electric arcs as they discharge.

[0004] Thus, there is a need for improved systems and methods that can be used to produce high quality devices and structures. These and other needs are addressed by the present technology.

SUMMARY

[0005] Exemplary semiconductor processing methods may include flowing a fluorine-containing precursor into a processing region of a semiconductor processing chamber. A substrate may be positioned within the processing region. The substrate may include alternating layers of silicon nitride and silicon oxide. The methods may include forming plasma effluents of the fluorine-containing precursor. The methods may include contacting the substrate with the plasma effluents of the fluorine-containing precursor. The contacting may selectively etch an exposed portion of silicon nitride. The methods may include introducing a phosphorous-and-fluorine-containing precursor into the processing region of the semiconductor processing chamber while maintaining a flow of the fluorine-containing precursor. The methods may include forming plasma effluents of the fluorine-containing precursor and the phosphorous-and-fluorine-containing precursor. The methods may include contacting the substrate with the plasma effluents of the fluorine-containing precursor and the phosphorous-and-fluorine-containing precursor. The contacting may selectively etch an exposed portion of silicon oxide.

[0006] In some embodiments, the fluorine-containing precursor may be or include hydrogen fluoride (HF). The phosphorous-and-fluorine-containing precursor may be or include phosphorous trifluoride (PF₃). A flow rate of the fluorine-containing precursor may be greater than or about 100 sccm. A flow rate of the phosphorous-and-fluorine-containing precursor may be less than or

about 50 sccm. The fluorine-containing precursor may be carbon-free. A source plasma power may be pulsed while forming the plasma effluents of the fluorine-containing precursor and the plasma effluents of the fluorine-containing precursor and the phosphorous-and-fluorine-containing precursor. The methods may include applying a bias power while contacting the stacked layers with the plasma effluents of the fluorine-containing precursor and the plasma effluents of the fluorine-containing precursor and the phosphorous-and-fluorine-containing precursor. The bias power may be greater than or about **100 W**. The bias power may be pulsed while contacting the stacked layers with the plasma effluents of the fluorine-containing precursor and the plasma effluents of the fluorine-containing precursor and the phosphorous-and-fluorine-containing precursor. The method may be performed at a chamber operating pressure of less than or about 500 mTorr. The method may be performed at a substrate operating temperature of less than or about 0° C.

[0007] Some embodiments of the present technology may encompass semiconductor processing methods. The methods may include i) flowing a fluorine-containing precursor into a processing region of a semiconductor processing chamber. A substrate may be positioned within the processing region. The substrate may include alternating layers of silicon nitride and silicon oxide. The substrate may include a patterned resist material overlying the alternating layers of silicon nitride and silicon oxide. The methods may include ii) forming plasma effluents of the fluorine-containing precursor. The methods may include iii) contacting the stacked layers with the plasma effluents of the fluorine-containing precursor. The contacting may selectively etch an exposed layer of silicon nitride. The methods may include iv) flowing a phosphorous-and-fluorine-containing precursor into the processing region of the semiconductor processing chamber with the fluorine-containing precursor. The methods may include v) forming plasma effluents of the fluorine-containing precursor and the phosphorous-and-fluorine-containing precursor. The methods may include vi) contacting the stacked layers with the plasma effluents of the fluorine-containing precursor and the phosphorous-and-fluorine-containing precursor. The contacting may selectively etch an exposed layer of silicon oxide. The methods may include vii) repeating operations i) through vi) for at least a second cycle.

[0008] In some embodiments, the fluorine-containing precursor may further include hydrogen. An etch rate of the exposed layer of silicon nitride may be greater than or about 250 nm/min. An etch rate of the exposed layer of silicon nitride may be greater than or about 400 nm/min. Operations i) through vi) may be repeated for at least ten cycles. The method may be performed at a substrate operating temperature of less than or about -20° C.

[0009] Some embodiments of the present technology may encompass semiconductor processing methods. The methods may include flowing a fluorine-containing precursor into a processing region of a semiconductor processing chamber. A substrate may be positioned within the processing region. The substrate may include alternating layers of silicon nitride and silicon oxide. The methods may include forming plasma effluents of the fluorine-containing precursor. The methods may include contacting the substrate with the plasma effluents of the fluorine-containing precursor. The contacting may etch an exposed portion of silicon nitride relative to silicon oxide at a selectivity greater than or about 5:1. The methods may include introducing a phosphorous-and-fluorine-containing precursor into the processing region of the semiconductor processing chamber while maintaining a flow of the fluorine-containing precursor. The methods may include forming plasma effluents of the fluorine-containing precursor and the phosphorous-and-fluorine-containing precursor. The methods may include contacting the substrate with the plasma effluents of the fluorine-containing precursor and the phosphorous-and-fluorine-containing precursor. The contacting may etch an exposed portion of silicon oxide relative to silicon nitride at a selectivity greater than or about 5:1.

[0010] In some embodiments, a flow rate ratio of the fluorine-containing precursor to the phosphorous-and-fluorine-containing precursor may be greater than or about 30:1.

[0011] Such technology may provide numerous benefits over conventional systems and techniques.

For example, the processes may etch stacked layers including alternating layers of silicon nitride and silicon oxide within semiconductor structures. Additionally, the processes may uniformly etch through alternating layers of silicon nitride and silicon oxide without the use of polymerizing gases and/or greenhouse gases. These and other embodiments, along with many of their advantages and features, are described in more detail in conjunction with the below description and attached figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] A further understanding of the nature and advantages of the disclosed technology may be realized by reference to the remaining portions of the specification and the drawings.

[0013] FIG. 1 shows a top plan view of one embodiment of an exemplary processing system according to some embodiments of the present technology.

[0014] FIG. 2 shows a schematic cross-sectional view of an exemplary processing chamber according to some embodiments of the present technology.

[0015] FIG. 3 shows exemplary operations in a method according to some embodiments of the present technology.

[0016] FIGS. 4A-4D show cross-sectional views of substrates being processed according to some embodiments of the present technology.

[0017] Several of the figures are included as schematics. It is to be understood that the figures are for illustrative purposes, and are not to be considered of scale unless specifically stated to be of scale. Additionally, as schematics, the figures are provided to aid comprehension and may not include all aspects or information compared to realistic representations, and may include exaggerated material for illustrative purposes.

[0018] In the appended figures, similar components and/or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a letter that distinguishes among the similar components. If only the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the letter.

DETAILED DESCRIPTION

[0019] In transitioning from 2D NAND to 3D NAND, many process operations are modified from horizontal to vertical operations. Additionally, as 3D NAND structures grow in the number of cells being formed, the aspect ratios of memory holes and other structures increase, sometimes dramatically. During 3D NAND processing, stacks of placeholder layers and dielectric materials may form the inter-electrode dielectric or integrated passive device (IPD) layers. These placeholder layers may have a variety of operations performed to place structures before fully removing the material and replacing it with metal. For example, one or more memory holes or trenches may be etched into the stacks of placeholder layers and dielectric materials prior to removing the placeholder layers.

[0020] Many conventional technologies utilize an etch process that passivates sidewalls of the memory holes or trenches. By passivating the sidewalls, a uniform profile of the memory holes or trenches may be maintained, and lateral etching may be minimized. However, formation of the passivation material on the sidewalls, which may be a polymeric material, may not be uniform throughout a depth of the memory hole or trench. Accordingly, conventional technologies may suffer from pattern loading and/or bending. Further, conventional technologies forming polymeric passivation material may deposit polymeric material on the wafer bevel, which may result in arcing.

[0021] The present technology overcomes these issues by performing an etch process using cyclic

exposure first to a fluorine-containing precursor and second to the fluorine-containing precursor with a phosphorous-and-fluorine-containing precursor. The etch process may be formed at a low temperature that increases directionality of the etch without the need for polymeric passivation material. Due to the high directionality of the etch, issues with memory hole or trench profile are reduced and/or eliminated. Additionally, arcing is mitigated since polymeric material is not being inadvertently deposited on the wafer bevel.

[0022] Although the remaining disclosure will routinely identify specific etching processes utilizing the disclosed technology, it will be readily understood that the systems and methods are equally applicable to deposition and cleaning processes as may occur in the described chambers. Accordingly, the technology should not be considered to be so limited as for use with etching processes or chambers alone. Moreover, although an exemplary chamber is described to provide foundation for the present technology, it is to be understood that the present technology can be applied to virtually any semiconductor processing chamber that may allow the single-chamber operations described.

[0023] FIG. 1 shows a top plan view of one embodiment of a processing system **10** of deposition, etching, baking, and/or curing chambers according to embodiments. The tool or processing system **10** depicted in FIG. 1 may contain a plurality of process chambers, **24a-d**, a transfer chamber **20**, a service chamber **26**, an integrated metrology chamber **28**, and a pair of load lock chambers **16a-b**. The process chambers may include any number of structures or components, as well as any number or combination of processing chambers.

[0024] To transport substrates among the chambers, the transfer chamber **20** may contain a robotic transport mechanism **22**. The transport mechanism **22** may have a pair of substrate transport blades **22a** attached to the distal ends of extendible arms **22b**, respectively. The blades **22a** may be used for carrying individual substrates to and from the process chambers. In operation, one of the substrate transport blades such as blade **22a** of the transport mechanism **22** may retrieve a substrate **W** from one of the load lock chambers such as chambers **16a-b** and carry substrate **W** to a first stage of processing, for example, a treatment process as described below in chambers **24a-d**. The chambers may be included to perform individual or combined operations of the described technology. For example, while one or more chambers may be configured to perform a deposition or etching operation, one or more other chambers may be configured to perform a pre-treatment operation and/or one or more post-treatment operations described. Any number of configurations are encompassed by the present technology, which may also perform any number of additional fabrication operations typically performed in semiconductor processing.

[0025] If the chamber is occupied, the robot may wait until the processing is complete and then remove the processed substrate from the chamber with one blade **22a** and may insert a new substrate with a second blade. Once the substrate is processed, it may then be moved to a second stage of processing. For each move, the transport mechanism **22** generally may have one blade carrying a substrate and one blade empty to execute a substrate exchange. The transport mechanism **22** may wait at each chamber until an exchange can be accomplished.

[0026] Once processing is complete within the process chambers, the transport mechanism **22** may move the substrate **W** from the last process chamber and transport the substrate **W** to a cassette within the load lock chambers **16a-b**. From the load lock chambers **16a-b**, the substrate may move into a factory interface **12**. The factory interface **12** generally may operate to transfer substrates between pod loaders **14a-d** in an atmospheric pressure clean environment and the load lock chambers **16a-b**. The clean environment in factory interface **12** may be generally provided through air filtration processes, such as HEPA filtration, for example. Factory interface **12** may also include a substrate orienter/aligner that may be used to properly align the substrates prior to processing. At least one substrate robot, such as robots **18a-b**, may be positioned in factory interface **12** to transport substrates between various positions/locations within factory interface **12** and to other locations in communication therewith. Robots **18a-b** may be configured to travel along a track

system within factory interface **12** from a first end to a second end of the factory interface **12**. [0027] The processing system **10** may further include an integrated metrology chamber **28** to provide control signals, which may provide adaptive control over any of the processes being performed in the processing chambers. The integrated metrology chamber **28** may include any of a variety of metrological devices to measure various film properties, such as thickness, roughness, composition, and the metrology devices may further be capable of characterizing grating parameters such as critical dimensions, sidewall angle, and feature height under vacuum in an automated manner.

[0028] Each of processing chambers **24a-d** may be configured to perform one or more process steps in the fabrication of a semiconductor structure, and any number of processing chambers and combinations of processing chambers may be used on multi-chamber processing system **10**. For example, any of the processing chambers may be configured to perform a number of substrate processing operations including any number of deposition processes including cyclical layer deposition, atomic layer deposition, chemical vapor deposition, physical vapor deposition, as well as other operations including etch, pre-clean, pre-treatment, post-treatment, anneal, plasma processing, degas, orientation, and other substrate processes. Some specific processes that may be performed in any of the chambers or in any combination of chambers may be metal deposition, surface cleaning and preparation, thermal annealing such as rapid thermal processing, and plasma processing. Any other processes may similarly be performed in specific chambers incorporated into multi-chamber processing system **10**, including any process described below, as would be readily appreciated by the skilled artisan.

[0029] FIG. **2** illustrates a schematic cross-sectional view of an exemplary processing chamber **100** suitable for patterning a material layer disposed on a substrate **302** in the processing chamber **100**. The exemplary processing chamber **100** is suitable for performing a patterning process, although it is to be understood that aspects of the present technology may be performed in any number of chambers, and substrate supports according to the present technology may be included in etching chambers, deposition chambers, treatment chambers, or any other processing chamber. The plasma processing chamber **100** may include a chamber body **105** defining a chamber volume **101** in which a substrate may be processed. The chamber body **105** may have sidewalls **112** and a bottom **118** which are coupled with ground **126**. The sidewalls **112** may have a liner **115** to protect the sidewalls **112** and extend the time between maintenance cycles of the plasma processing chamber **100**. The dimensions of the chamber body **105** and related components of the plasma processing chamber **100** are not limited and generally may be proportionally larger than the size of the substrate **302** to be processed therein. Examples of substrate sizes include 200 mm diameter, 250 mm diameter, 300 mm diameter and 450 mm diameter, among others, such as display or solar cell substrates as well.

[0030] The chamber body **105** may support a chamber lid assembly **110** to enclose the chamber volume **101**. The chamber body **105** may be fabricated from aluminum or other suitable materials. A substrate access port **113** may be formed through the sidewall **112** of the chamber body **105**, facilitating the transfer of the substrate **302** into and out of the plasma processing chamber **100**. The access port **113** may be coupled with a transfer chamber and/or other chambers of a substrate processing system as previously described. A pumping port **145** may be formed through the sidewall **112** of the chamber body **105** and connected to the chamber volume **101**. A pumping device may be coupled through the pumping port **145** to the chamber volume **101** to evacuate and control the pressure within the processing volume. The pumping device may include one or more pumps and throttle valves.

[0031] A gas panel **160** may be coupled by a gas line **167** with the chamber body **105** to supply process gases into the chamber volume **101**. The gas panel **160** may include one or more process gas sources **161**, **162**, **163**, **164** and may additionally include inert gases, non-reactive gases, and reactive gases, as may be utilized for any number of processes. Examples of process gases that may

be provided by the gas panel **160** include, but are not limited to, hydrocarbon containing gas including methane, sulfur hexafluoride, silicon chloride, carbon tetrafluoride, hydrogen bromide, hydrocarbon containing gas, argon gas, chlorine, nitrogen, helium, or oxygen gas, as well as any number of additional materials. Additionally, process gasses may include nitrogen, chlorine, fluorine, oxygen, and hydrogen containing gases such as H.sub.2, NH.sub.3, H.sub.2O, H.sub.2O.sub.2, NF.sub.3, HF, F.sub.2, CF.sub.4, CHF.sub.3, C.sub.2F.sub.6, C.sub.2F.sub.4, C.sub.3F.sub.6, C.sub.4F.sub.6, C.sub.4F.sub.8, BrF.sub.3, ClF.sub.3, SF.sub.6, CH.sub.3F, CH.sub.2F.sub.2, BC.sub.13, PF.sub.3, PH.sub.3, COS, and SO.sub.2, among any number of additional precursors.

[0032] Valves **166** may control the flow of the process gases from the sources **161, 162, 163, 164** from the gas panel **160** and may be managed by a controller **165**. The flow of the gases supplied to the chamber body **105** from the gas panel **160** may include combinations of the gases from one or more sources. The lid assembly **110** may include a nozzle **114**. The nozzle **114** may be one or more ports for introducing the process gases from the sources **161, 162, 164, 163** of the gas panel **160** into the chamber volume **101**. After the process gases are introduced into the plasma processing chamber **100**, the gases may be energized to form plasma. An antenna **148**, such as one or more inductor coils, may be provided adjacent to the plasma processing chamber **100**. An antenna power supply **142** may power the antenna **148** through a match circuit **141** to inductively couple energy, such as RF energy, to the process gas to maintain a plasma formed from the process gas in the chamber volume **101** of the plasma processing chamber **100**. Alternatively, or in addition to the antenna power supply **142**, process electrodes below the substrate **302** and/or above the substrate **302** may be used to capacitively couple RF power to the process gases to maintain the plasma within the chamber volume **101**. The operation of the power supply **142** may be controlled by a controller, such as controller **165**, that also controls the operation of other components in the plasma processing chamber **100**.

[0033] A substrate support pedestal **135** may be disposed in the chamber volume **101** to support the substrate **302** during processing. The substrate support pedestal **135** may include an electrostatic chuck ("ESC") **122** for holding the substrate **302** during processing. The electrostatic chuck **122** may use the electrostatic attraction to hold the substrate **302** to the substrate support pedestal **135**. The ESC **122** may be powered by an RF power supply **125** integrated with a match circuit **124**. The ESC **122** may include an electrode **121** embedded within a dielectric body. The electrode **121** may be coupled with the RF power supply **125** and may provide a bias which attracts plasma ions, formed by the process gases in the chamber volume **101**, to the ESC **122** and substrate **302** seated on the pedestal. The RF power supply **125** may cycle on and off, or pulse, during processing of the substrate **302**. The ESC **122** may have an isolator **128** for the purpose of making the sidewall of the ESC **122** less attractive to the plasma to prolong the maintenance life cycle of the ESC **122**. Additionally, the substrate support pedestal **135** may have a cathode liner **136** to protect the sidewalls of the substrate support pedestal **135** from the plasma gases and to extend the time between maintenance of the plasma processing chamber **100**.

[0034] Electrode **121** may be coupled with a power source **150**. The power source **150** may provide a chucking voltage of about 500 volts to about 15,000 volts to the electrode **121**. The power source **150** may also include a system controller for controlling the operation of the electrode **121** by directing a DC current to the electrode **121** for chucking and de-chucking the substrate **302**. For example, similar to the RF power supply **125**, power supply **150** may provide a bias which attracts plasma ions, formed by the process gases in the chamber volume **101**, to the ESC **122** and substrate **302** seated on the pedestal. The power supply **150** may cycle on and off, or pulse, during processing of the substrate **302**. In embodiments, the power supply **150** may supply RF power, DC current or voltage for chucking and/or bias, or a combination thereof. In additional embodiments, multiple power supplies may be configured to supply RF power and DC current or voltage for chucking and/or bias. The ESC **122** may include heaters disposed within the pedestal and

connected to a power source for heating the substrate, while a cooling base **129** supporting the ESC **122** may include conduits for circulating a heat transfer fluid to maintain a temperature of the ESC **122** and substrate **302** disposed thereon. The ESC **122** may be configured to perform in the temperature range required by the thermal budget of the device being fabricated on the substrate **302**. For example, the ESC **122** may be configured to maintain the substrate **302** at a temperature of about -150°C . or lower to about 500°C . or higher depending on the process being performed. [0035] The cooling base **129** may be provided to assist in controlling the temperature of the substrate **302**. To mitigate process drift and time, the temperature of the substrate **302** may be maintained substantially constant by the cooling base **129** throughout the time the substrate **302** is in the cleaning chamber. In some embodiments, the temperature of the substrate **302** may be maintained throughout subsequent cleaning processes at temperatures between about -150°C . and about 500°C ., although any temperatures may be utilized. A cover ring **130** may be disposed on the ESC **122** and along the periphery of the substrate support pedestal **135**. The cover ring **130** may be configured to confine etching gases to a desired portion of the exposed top surface of the substrate **302**, while shielding the top surface of the substrate support pedestal **135** from the plasma environment inside the plasma processing chamber **100**. Lift pins may be selectively translated through the substrate support pedestal **135** to lift the substrate **302** above the substrate support pedestal **135** to facilitate access to the substrate **302** by a transfer robot or other suitable transfer mechanism as previously described.

[0036] The controller **165** may be utilized to control the process sequence, regulating the gas flows from the gas panel **160** into the plasma processing chamber **100**, and other process parameters. Software routines, when executed by the CPU, transform the CPU into a specific purpose computer such as a controller, which may control the plasma processing chamber **100** such that the processes are performed in accordance with the present disclosure. The software routines may also be stored and/or executed by a second controller that may be associated with the plasma processing chamber **100**.

[0037] The chamber discussed previously may be used in performing exemplary methods, including etching methods, although any number of chambers may be configured to perform one or more aspects used in embodiments of the present technology. Turning to FIG. **3**, exemplary operations in a method **300** according to embodiments of the present technology are shown. Method **300** may include one or more operations prior to the initiation of the method, including front end processing, deposition, etching, polishing, cleaning, or any other operations that may be performed prior to the described operations. The methods may include a number of optional operations, which may or may not be specifically associated with some embodiments of methods, according to embodiments of the present technology. For example, many of the operations are described in order to provide a broader scope of the processes performed, but are not critical to the technology, or may be performed by alternative methodology as will be discussed further below. Method **300** may describe operations shown schematically in FIGS. **4A-4D**, the illustrations of which will be described in conjunction with the operations of method **300**. It is to be understood that the figures illustrate only partial schematic views, and a substrate may contain any number of additional materials and features having a variety of characteristics and aspects as illustrated in the figures.

[0038] Method **300** may or may not involve optional operations to develop the semiconductor structure to a particular fabrication operation. It is to be understood that method **300** may be performed on any number of semiconductor structures **500** or substrates **405**, as illustrated in FIG. **4A**, including exemplary structures on which a silicon oxide and silicon nitride etching operation may be performed. As illustrated in FIG. **4A**, substrate **405** may have a plurality of stacked layers overlying the substrate, which may be silicon, silicon germanium, or other substrate materials. The layers may include IPD layers including dielectric material **410**, which may be silicon oxide, in alternating layers with placeholder material **420**, which may be silicon nitride or polysilicon, for

example. Placeholder material **420** may be or include material that will be removed to produce individual memory cells in subsequent operations. Although the remaining disclosure will discuss silicon nitride and silicon oxide IPD layers, any other known materials used in these two layers may be substituted for one or more of the layers. Although illustrated with only 7 layers of material, exemplary structures may include any number of layers including hundreds of layers of material, and it is to be understood that the figures are only schematics to illustrate aspects of the present technology. Additionally, to allow for one or more memory holes or trenches to be formed through the IPD layers, a mask material **425** may be formed on the alternating layers of the dielectric material **410** and the placeholder material **420**. The mask material **425**, such as resist material, may be patterned to form one or more apertures **430**, exposing a portion of the underlying alternating layers. Although only a single aperture **430** is illustrated, it is to be understood that exemplary structure **400** may include any number of apertures across the substrate **405**. Some or all of these operations may be performed in chambers or system tools as previously described, or may be performed in different chambers on the same system tool, which may include the chamber in which the operations of method **300** are performed.

[0039] Method **300** may be performed to etch or otherwise remove portions of the dielectric material **410** and the placeholder material **420**, which may form memory holes or trenches in the structure **400** as illustrated. The method may be performed to facilitate control of the profile through the structure, and improve etch characteristics, such as uniformity of the memory holes or trenches as the etch progresses into the alternating layers. Method **300** may include flowing a fluorine-containing precursor into a processing region of the semiconductor processing chamber in which the substrate is maintained at operation **305**. Plasma effluents of the fluorine-containing precursor may be formed at operation **315**. The plasma effluents of the fluorine-containing precursor may contact the substrate **405** at operation **310**, and may selectively etch an exposed portion, such as material exposed through the aperture **430** in the mask material **425**, of the placeholder material **420**. For example, the first material exposed through the aperture **430** may be placeholder material **420**. The plasma effluents of the fluorine-containing precursor may selectively etch the placeholder material **420** to expose an underlying layer of dielectric material **410**.

[0040] In embodiments, method **300** may include introducing a phosphorous-and-fluorine-containing precursor into the processing region of the semiconductor processing chamber while maintaining a flow of the fluorine-containing precursor at operation **320**. Plasma effluents of the phosphorous-and-fluorine-containing precursor, as well as the fluorine-containing precursor, may be formed at operation **325**. The plasma effluents of the phosphorous-and-fluorine-containing precursor, as well as the fluorine-containing precursor, may contact the substrate **405** at operation **330**, and may selectively etch an exposed portion of dielectric material at operation **330**. The plasma effluents of the phosphorous-and-fluorine-containing precursor and the fluorine-containing precursor may selectively etch the dielectric material **410** to expose an underlying layer of placeholder material **420**. The phosphorous-and-fluorine-containing precursor may increase the etch rate of oxide material, such as dielectric material **410**, with minor impact on the nitride etch rate, such as placeholder material **420**. Accordingly, method **300** may toggle the flow of the phosphorous-and-fluorine-containing precursor to selectively etch oxide material, such as dielectric material **410**. As illustrated in FIG. **4B**, the alternative etching of dielectric material **410** and placeholder material **420** may form a memory hole or trench **435** in the stacked layers. The memory hole or trench **435** may be in alignment with the aperture **430** in the mask material **425**. Again, although only a single memory hole or trench **435** is illustrated, it is to be understood that exemplary structure **400** may include any number of memory holes or trenches across the substrate **405**.

[0041] As illustrated in FIGS. **4C-4D**, the operations of method **300** may be repeated for a second cycle and may be repeated for any number of cycles. The number of cycles may be dependent on a desired depth of the memory hole or trench **435**, which may be based on the number of alternating

layers of the dielectric material **410** and the placeholder material **420**. In embodiments, the depth of the memory hole or trench **435** may extend through all of the stacked layers of the dielectric material **410** and the placeholder material **420**, which may ultimately expose the substrate **405**.

[0042] Fluorine-containing precursors flowed at operation **305** and during operation **320** may include hydrogen fluoride (HF), nitrogen trifluoride (NF₃), diatomic fluorine (F₂), bromine trifluoride (BrF₃), chlorine trifluoride (ClF₃), sulfur hexafluoride (SF₆), xenon difluoride (XeF₂), or any other fluorine-containing precursor used or useful in semiconductor processing. In some embodiments, the fluorine-containing precursor may include hydrogen. In some embodiments, the fluorine-containing precursor may be carbon-free. For example, the fluorine-containing precursor may be HF. In embodiments, the fluorine-containing precursor may be free of polymerizing gases and/or greenhouse gases. The fluorine-containing precursor may also be flowed with any number of additional precursors or carrier gases including nitrogen, argon, helium, or any number of additional materials, although in some embodiments the precursors may be limited to control side reactions or other aspects that may impact the etching.

[0043] Phosphorus-and-fluorine-containing precursors flowed at operation **320** may include phosphorus trifluoride (PF₃), diphosphorus tetrafluoride (P₂F₄), or any other phosphorus-and-fluorine-containing precursor used or useful in semiconductor processing. In embodiments, the phosphorus-and-fluorine-containing precursor may be free of polymerizing gases and/or greenhouse gases. As such, in some embodiments, the phosphorus-and-fluorine-containing precursor may be carbon-free. The phosphorus-and-fluorine-containing precursor may also be flowed with any number of additional precursors or carrier gases including nitrogen, argon, helium, or any number of additional materials, although in some embodiments the precursors may be limited to control side reactions or other aspects that may impact the etching.

[0044] A flow rate of the fluorine-containing precursor at operation **305** and/or operation **320** may be greater than or about 100 sccm, and may be greater than or about 125 sccm, greater than or about 150 sccm, greater than or about 175 sccm, greater than or about 200 sccm, greater than or about 225 sccm, greater than or about 250 sccm, greater than or about 260 sccm, greater than or about 270 sccm, greater than or about 280 sccm, greater than or about 290 sccm, greater than or about 300 sccm, or more. At flow rates less than, for example, 100 sccm, reduced fluorine may be present and may result in slower etching and reduced throughput. However, excessive fluorine may result in lateral etching that may impact uniformity of the etch. As such, the flow rate of the fluorine-containing precursor at operation **305** and/or operation **320** may be less than or about 500 sccm, and may be less than or about 450 sccm, less than or about 400 sccm, less than or about 350 sccm, less than or about 300 sccm, or less. The flow rate of the fluorine-containing precursor may be constant between operation **305** and operation **320**. However, it is also contemplated that the flow rate of the fluorine-containing precursor may be adjusted between operation **305** and operation **320**.

[0045] A flow rate of the phosphorus-and-fluorine-containing precursor at operation **320** may be less than or about 100 sccm, and may be less than or about 90 sccm, less than or about 80 sccm, less than or about 70 sccm, less than or about 60 sccm, less than or about 50 sccm, less than or about 40 sccm, less than or about 30 sccm, less than or about 20 sccm, less than or about 10 sccm, less than or about 9 sccm, less than or about 8 sccm, less than or about 7 sccm, less than or about 6 sccm, less than or about 5 sccm, less than or about 4 sccm, less than or about 3 sccm, or less. At increased flow rates, an etch rate of the dielectric material **410** may increase due to the presence of phosphorous in the dielectric material **410**, such as silicon oxide. However, excessive phosphorous may not benefit the etch rate.

[0046] A flow rate ratio of the fluorine-containing precursor to the phosphorous-and-fluorine-containing precursor may be adjusted to control the amount of phosphorus present in the plasma effluents. At reduced flow rate ratios of the fluorine-containing precursor to the phosphorous-and-fluorine-containing precursor, additional phosphorous may interact with the dielectric material **410**,

such as silicon oxide, which may increase the etch rate of the dielectric material **410**. The phosphorous may enter the silicon and oxygen lattice and increase the etch rate of the dielectric material **410**. With too much phosphorous in the plasma effluents, the etch rate may be too high and precise control of the etch may be reduced. Accordingly, the flow rate ratio of the fluorine-containing precursor to the phosphorous-and-fluorine-containing precursor may be greater than or about 30:1, and may be greater than or about 35:1, greater than or about 40:1, greater than or about 45:1, greater than or about 50:1, greater than or about 55:1, greater than or about 60:1, greater than or about 65:1, greater than or about 70:1, greater than or about 75:1, greater than or about 80:1, greater than or about 85:1, greater than or about 90:1, or more. However, at very high flow rate ratios of the fluorine-containing precursor to the phosphorous-and-fluorine-containing precursor, the effect of phosphorous on the etching of the dielectric material **410** may be reduced.

[0047] A source plasma power used to form plasma effluents of the fluorine-containing precursor and/or the phosphorous-and-fluorine-containing precursor and fluorine-containing precursor may be a relatively high plasma power. The relatively high plasma power may allow for formation of fluorine radicals. At lower plasma powers, the phosphorous-and-fluorine-containing precursor and fluorine-containing precursor may suffer from reduced dissociation and the etch amount per cycle may reduce. Accordingly, the plasma effluents of the fluorine-containing precursor and/or the phosphorous-and-fluorine-containing precursor and fluorine-containing precursor may be formed at greater than or about 500 W, and may be formed at greater than or about 750 W, greater than or about 1,000 W, greater than or about 1,250 W, greater than or about 1,500 W, greater than or about 1,750 W, greater than or about 2,000 W, greater than or about 2,250 W, greater than or about 2,500 W, greater than or about 2,750 W, greater than or about 3,000 W, greater than or about 3,250 W, greater than or about 3,500 W, greater than or about 3,750 W, greater than or about 4,000 W, or more. However, very high plasma powers may result in damage to the structure **400**. Therefore, the plasma effluents of the fluorine-containing precursor and/or the phosphorous-and-fluorine-containing precursor and fluorine-containing precursor may be formed at less than or about 6,000 W, and may be formed at greater than or about 5,500 W, less than or about 5,000 W, less than or about 4,500 W, less than or about 4,000 W, or less. The plasma power used to form plasma effluents of the fluorine-containing precursor at operation **310** may be the same or different than the plasma power used to form plasma effluents of the fluorine-containing precursor and the phosphorous-and-fluorine-containing precursor at operation **325**.

[0048] In embodiments, the source plasma power may be pulsed while forming the plasma effluents of the fluorine-containing precursor, while forming plasma effluents of the plasma effluents of the fluorine-containing precursor and the phosphorous-and-fluorine-containing precursor, or both. A duty cycle of the source plasma power may be applied at less than or about 95%, and may be applied at less than or about 90%, less than or about 85%, to reduce the effective plasma power.

[0049] As previously discussed, the contacting at operation **315** with plasma effluents of the fluorine-containing precursor may selectively etch an exposed portion of the placeholder material **420**. The contacting at operation **315** may etch an exposed portion of placeholder material **420** relative to dielectric material **410** at a selectivity greater than or about 5:1, such as a selectivity of greater than or about 7:1, greater than or about 8:1, greater than or about 9:1, greater than or about 10:1, or more. In embodiments, an etch rate of the exposed placeholder material **420** may be greater than or about 250 nm/min, and may be greater than or about 300nm/min, greater than or about 350 nm/min, greater than or about 400 nm/min, greater than or about 450 nm/min, greater than or about 500 nm/min, greater than or about 550 nm/min, or more.

[0050] Similarly, the contacting at operation **325** with plasma effluents of the phosphorous-and-fluorine-containing precursor and the fluorine-containing precursor may selectively etch an exposed portion of the dielectric material **410**. The contacting at operation **325** may etch an exposed portion of dielectric material **410** relative to placeholder material **420** at a selectivity

greater than or about 5:1, such as a selectivity of greater than or about 7:1, greater than or about 8:1, greater than or about 9:1, greater than or about 10:1, or more. In embodiments, an etch rate of the exposed placeholder material **420** may be greater than or about 350 nm/min, and may be greater than or about 400 nm/min, greater than or about 450 nm/min, greater than or about 500 nm/min, greater than or about 550 nm/min, greater than or about 600 nm/min, greater than or about 650 nm/min, greater than or about 700 nm/min, or more.

[0051] While forming the plasma effluents at operation **310**, operation **325**, and/or while contacting the substrate with the plasma effluents of the plasma effluent at operations **315** and/or **330**, a bias power may be applied. The bias power may be provided through an RF power source, such as a bias power source, or as a voltage through the pedestal or substrate support. In embodiment, the bias power may be provided at a 2 MHz frequency. The bias power may increase directionality of the plasma effluents of the inert precursor. The increased directionality may draw the plasma effluents to the substrate **405**. Accordingly, the plasma effluents may bombard and remove the exposed portion of the stacked layers of the dielectric material **410** and the placeholder material **420**. In embodiments, the bias power applied may be greater than or about 100 W, and may be formed at greater than or about 200 W, greater than or about 300 W, greater than or about 400 W, greater than or about 500 W, greater than or about 600 W, greater than or about 700 W, greater than or about 750 W, greater than or about 800 W, greater than or about 900 W, greater than or about 1,000 W, or more. However, at higher bias powers, the bombardment may increase and materials on substrate **405** or in structure **400** may begin to sputter. Accordingly, the bias power applied may be less than or about 5,500 W, less than or about 5,000 W, less than or about 4,500 W, less than or about 4,000 W, or less. In embodiments, the bias power may be pulsed or have a non-continuous power on time. For example, the power on time of the bias power may be less than or about 75%, and may be less than or about 50%, less than or about 45%, less than or about 40%, less than or about 35%, less than or about 30%, less than or about 25%, less than or about 20%, or less.

[0052] By performing an amount of removal or etch of nitride material, such as placeholder material **420**, followed by an amount of removal or etch of oxide material, such as dielectric material **410**, a controlled directional etch of the stacked layers of the dielectric material **410** and the placeholder material **420** may be performed. As illustrated in FIGS. **4C-4D**, to further facilitate directional etching, the present technology may be performed in a number of cycles of providing the fluorine-containing precursor and providing the phosphorous-and-fluorine-containing precursor with the fluorine-containing precursor to allow efficient etching of the stacked layers of the dielectric material **410** and the placeholder material **420**. In some embodiments, method **300** may include repeating the operations **305-330** for at least two cycles, and may include repeating the operations **305-335** for at least three cycles, at least four cycles, at least five cycles, at least ten cycles, at least fifteen cycles, at least twenty cycles, or more. It is contemplated that the operations **305-330** of method **300** may be repeated any number of times depending on depth of the memory hole or trench **435** to be etched. Additionally, while the preceding description discusses FIGS. **4A-4D** depict the etching of placeholder material **420** occurring first, it is contemplated that the method may occur with the etching of dielectric material **410** first.

[0053] Process conditions may also impact the operations performed in method **300**. Each of the operations of method **300** may be performed during a constant temperature in embodiments, while in some embodiments the temperature may be adjusted during different operations.

[0054] Temperatures may be maintained in any range, however, at higher temperatures, directionality of the plasma effluents may decrease. As directionality decreases, profile control of the memory holes or trench **435** being etched may suffer. Additionally, at lower temperatures, water (H.sub.2O), a byproduct from the removal of silicon oxide, may condense on the stacked layers of the dielectric material **410** and the placeholder material **420** and serve to accelerate the etch.

[0055] Accordingly, in some embodiments any or all operations of the method **300** may performed at a substrate operating temperature of less than or about 20° C., and may be performed at a

substrate operating temperature of less than or about 10° C., less than or about 0° C., less than or about -10° C., less than or about -20° C., less than or about -30° C., less than or about -40° C., less than or about -50° C., less than or about -60° C., less than or about -70° C., less than or about -80° C., or less.

[0056] Each of the operations of method **300** may be performed during a constant pressure in embodiments, while in some embodiments the pressure may be adjusted during different operations. Pressures may be maintained in any range, however, at higher pressures, further dissociation of the fluorine-containing precursors may occur, which may produce more fluorine radicals. As the amount of fluorine radicals increases, directionality of the etch may decrease and profile control of the memory hole or trench **435** may suffer. Accordingly, in some embodiments any or all operations of the method **300** may be performed at a chamber operating pressure of greater than or about 5 mTorr, and may be performed at a chamber operating pressure of greater than or about 10 mTorr, greater than or about 15 mTorr, greater than or about 20 mTorr, greater than or about 25 mTorr, greater than or about 30 mTorr, greater than or about 35 mTorr, greater than or about 40 mTorr, greater than or about 45 mTorr, greater than or about 50 mTorr, greater than or about 55 mTorr, greater than or about 60 mTorr, greater than or about 75 mTorr, greater than or about 100 mTorr, greater than or about 200 mTorr, greater than or about 300 mTorr, greater than or about 400 mTorr, or more. Conversely, at lower pressures, directionality of the etch may increase as well as ion energy and etch amount per cycle. Therefore, any or all operations of the method **300** may be performed at a chamber operating pressure of less than or about 500 mTorr, and may be performed at a chamber operating pressure of less than or about 400 mTorr, less than or about 300 mTorr, less than or about 200 mTorr, less than or about 100 mTorr, less than or about 90 mTorr, less than or about 80 mTorr, less than or about 70 mTorr, less than or about 65 mTorr, less than or about 60 mTorr, less than or about 55 mTorr, less than or about 50 mTorr, less than or about 45 mTorr, less than or about 40 mTorr, less than or about 35 mTorr, less than or about 30 mTorr, less than or about 25 mTorr, less than or about 20 mTorr, less than or about 15 mTorr, less than or about 10 mTorr, less than or about 5 mTorr, or less.

[0057] In the preceding description, for the purposes of explanation, numerous details have been set forth in order to provide an understanding of various embodiments of the present technology. It will be apparent to one skilled in the art, however, that certain embodiments may be practiced without some of these details, or with additional details.

[0058] Having disclosed several embodiments, it will be recognized by those of skill in the art that various modifications, alternative constructions, and equivalents may be used without departing from the spirit of the embodiments. Additionally, a number of well-known processes and elements have not been described in order to avoid unnecessarily obscuring the present technology.

Accordingly, the above description should not be taken as limiting the scope of the technology. Additionally, methods or processes may be described as sequential or in steps, but it is to be understood that the operations may be performed concurrently, or in different orders than listed.

[0059] Where a range of values is provided, it is understood that each intervening value, to the smallest fraction of the unit of the lower limit, unless the context clearly dictates otherwise, between the upper and lower limits of that range is also specifically disclosed. Any narrower range between any stated values or unstated intervening values in a stated range and any other stated or intervening value in that stated range is encompassed. The upper and lower limits of those smaller ranges may independently be included or excluded in the range, and each range where either, neither, or both limits are included in the smaller ranges is also encompassed within the technology, subject to any specifically excluded limit in the stated range. Where the stated range includes one or both of the limits, ranges excluding either or both of those included limits are also included.

[0060] As used herein and in the appended claims, the singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise. Thus, for example, reference to “a precursor” includes a plurality of such precursors, and reference to “the layer” includes reference to

one or more layers and equivalents thereof known to those skilled in the art, and so forth. [0061] Also, the words “comprise(s)”, “comprising”, “contain(s)”, “containing”, “include(s)”, and “including”, when used in this specification and in the following claims, are intended to specify the presence of stated features, integers, components, or operations, but they do not preclude the presence or addition of one or more other features, integers, components, operations, acts, or groups.

Claims

1. A semiconductor processing method comprising: flowing a fluorine-containing precursor into a processing region of a semiconductor processing chamber, wherein a substrate is positioned within the processing region, and wherein the substrate comprises alternating layers of silicon nitride and silicon oxide; forming plasma effluents of the fluorine-containing precursor; contacting the substrate with the plasma effluents of the fluorine-containing precursor, wherein the contacting selectively etches an exposed portion of silicon nitride; introducing a phosphorous-and-fluorine-containing precursor into the processing region of the semiconductor processing chamber while maintaining a flow of the fluorine-containing precursor; forming plasma effluents of the fluorine-containing precursor and the phosphorous-and-fluorine-containing precursor; and contacting the substrate with the plasma effluents of the fluorine-containing precursor and the phosphorous-and-fluorine-containing precursor, wherein the contacting selectively etches an exposed portion of silicon oxide.
2. The semiconductor processing method of claim 1, wherein the fluorine-containing precursor comprises hydrogen fluoride (HF).
3. The semiconductor processing method of claim 1, wherein the phosphorous-and-fluorine-containing precursor comprises phosphorous trifluoride (PF₃).
4. The semiconductor processing method of claim 1, wherein a flow rate of the fluorine-containing precursor is greater than or about 100 sccm.
5. The semiconductor processing method of claim 1, wherein a flow rate of the phosphorous-and-fluorine-containing precursor is less than or about 50 sccm.
6. The semiconductor processing method of claim 1, wherein the fluorine-containing precursor is carbon-free.
7. The semiconductor processing method of claim 1, wherein a source plasma power is pulsed while forming the plasma effluents of the fluorine-containing precursor and the plasma effluents of the fluorine-containing precursor and the phosphorous-and-fluorine-containing precursor.
8. The semiconductor processing method of claim 1, further comprising: applying a bias power while contacting the stacked layers with the plasma effluents of the fluorine-containing precursor and the plasma effluents of the fluorine-containing precursor and the phosphorous-and-fluorine-containing precursor.
9. The semiconductor processing method of claim 8, wherein the bias power is greater than or about 100 W.
10. The semiconductor processing method of claim 8, wherein the bias power is pulsed while contacting the stacked layers with the plasma effluents of the fluorine-containing precursor and the plasma effluents of the fluorine-containing precursor and the phosphorous-and-fluorine-containing precursor.
11. The semiconductor processing method of claim 1, wherein the method is performed at a chamber operating pressure of less than or about 500 mTorr.
12. The semiconductor processing method of claim 1, wherein the method is performed at a substrate operating temperature of less than or about 0° C.
13. A semiconductor processing method comprising: i) flowing a fluorine-containing precursor into a processing region of a semiconductor processing chamber, wherein a substrate is positioned

within the processing region, wherein the substrate comprises alternating layers of silicon nitride and silicon oxide, and wherein the substrate comprises a patterned resist material overlying the alternating layers of silicon nitride and silicon oxide; ii) forming plasma effluents of the fluorine-containing precursor; iii) contacting the stacked layers with the plasma effluents of the fluorine-containing precursor, wherein the contacting selectively etches an exposed layer of silicon nitride; iv) flowing a phosphorous-and-fluorine-containing precursor into the processing region of the semiconductor processing chamber with the fluorine-containing precursor; v) forming plasma effluents of the fluorine-containing precursor and the phosphorous-and-fluorine-containing precursor; vi) contacting the stacked layers with the plasma effluents of the fluorine-containing precursor and the phosphorous-and-fluorine-containing precursor, wherein the contacting selectively etches an exposed layer of silicon oxide; and vii) repeating operations i) through vi) for at least a second cycle.

14. The semiconductor processing method of claim 13, wherein the fluorine-containing precursor further comprises hydrogen.

15. The semiconductor processing method of claim 13, wherein an etch rate of the exposed layer of silicon nitride is greater than or about 250 nm/min.

16. The semiconductor processing method of claim 13, wherein an etch rate of the exposed layer of silicon nitride is greater than or about 400 nm/min.

17. The semiconductor processing method of claim 13, operations i) through vi) are repeated for at least ten cycles.

18. The semiconductor processing method of claim 13, wherein the method is performed at a substrate operating temperature of less than or about -20°C .

19. A semiconductor processing method comprising: flowing a fluorine-containing precursor into a processing region of a semiconductor processing chamber, wherein a substrate is positioned within the processing region, and wherein the substrate comprises alternating layers of silicon nitride and silicon oxide; forming plasma effluents of the fluorine-containing precursor; contacting the substrate with the plasma effluents of the fluorine-containing precursor, wherein the contacting etches an exposed portion of silicon nitride relative to silicon oxide at a selectivity greater than or about 5:1; introducing a phosphorous-and-fluorine-containing precursor into the processing region of the semiconductor processing chamber while maintaining a flow of the fluorine-containing precursor; forming plasma effluents of the fluorine-containing precursor and the phosphorous-and-fluorine-containing precursor, and contacting the substrate with the plasma effluents of the fluorine-containing precursor and the phosphorous-and-fluorine-containing precursor, wherein the contacting etches an exposed portion of silicon oxide relative to silicon nitride at a selectivity greater than or about 5:1.

20. The semiconductor processing method of claim 19, wherein a flow rate ratio of the fluorine-containing precursor to the phosphorous-and-fluorine-containing precursor is greater than or about 30:1.
